

Module - III

Centroid & Moment of Inertia

Centre of Gravity

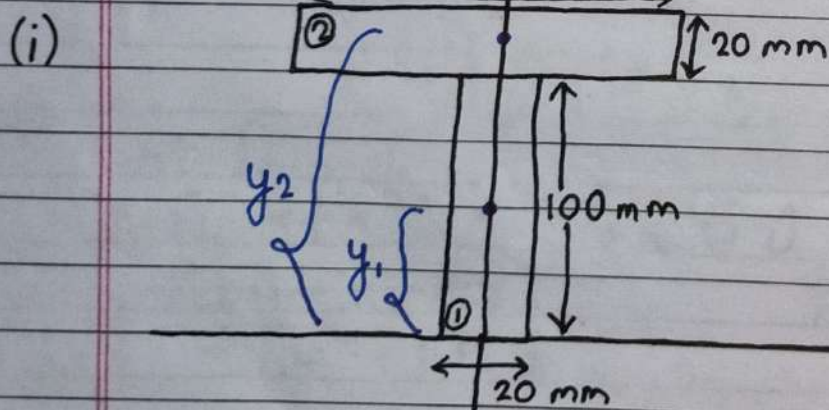
The centre of gravity of a body can be defined as the point through which the resultant of the force of the gravity of the body acts.

Difference Between Centre of Gravity & Centroid

The term "centre of gravity" applies to bodies with weight and centroid applies to area.

Numericals

Q. Find the centroid of the following:



Solⁿ:

$$\bar{x} = 0$$
$$\bar{y} = ?$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2}, \quad \bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$$\text{Area, } A_1 = A_2 = 100 \times 20 = 2000 \text{ mm}^2$$

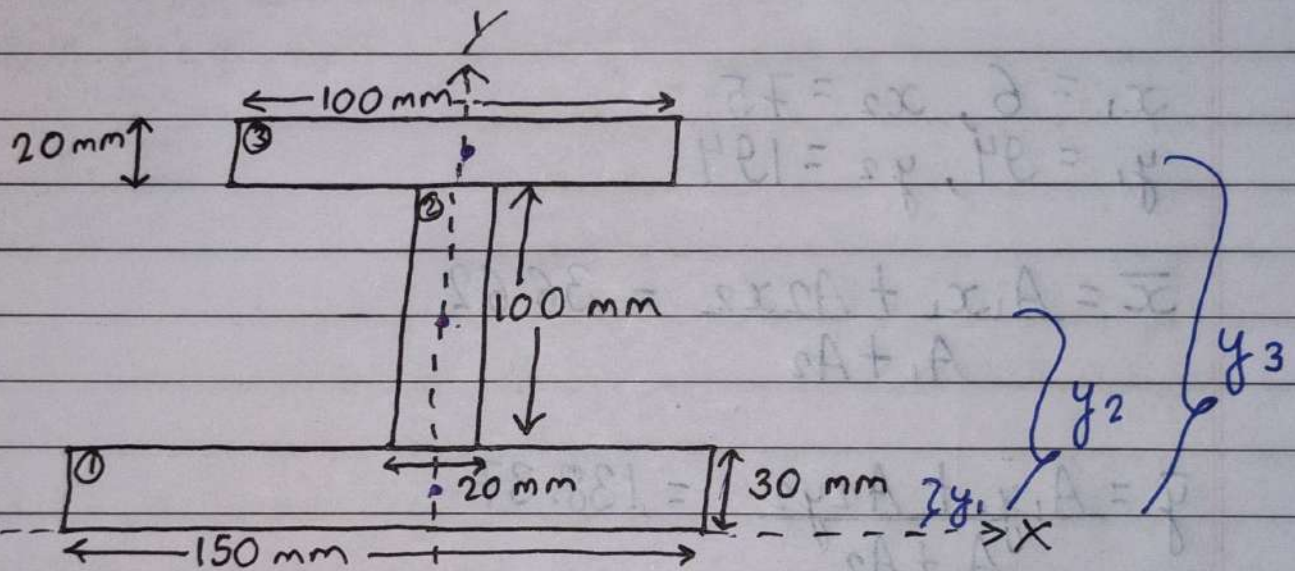
$$y_1 = 50 \text{ mm}$$

$$\& y_2 = 110 \text{ mm}$$

$$\therefore (\bar{x}, \bar{y}) = \left(0, \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} \right) = \left(0, \frac{2000 \times 50 + 2000 \times 110}{4000} \right)$$

$$\text{i.e. Centroid} = (\bar{x}, \bar{y}) = (0, 80).$$

(ii)



Solⁿ:

$$A_1 = 150 \times 30 = 4500 \text{ mm}^2$$

$$A_2 = A_3 = 100 \times 20 = 2000 \text{ mm}^2$$

$$y_1 = 15 \text{ mm}$$

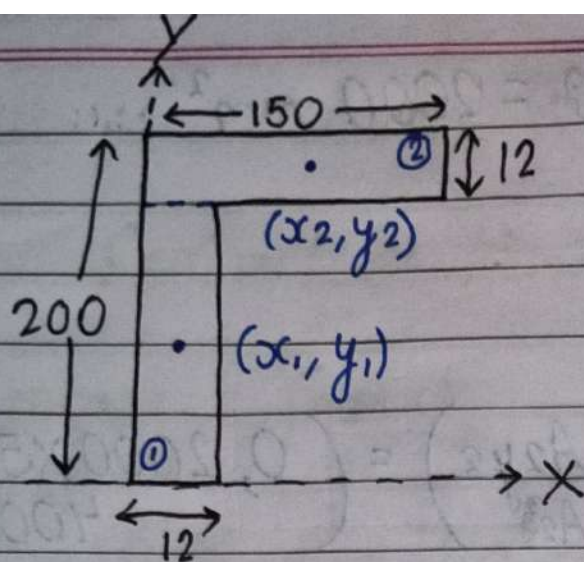
$$y_2 = 80 \text{ mm}$$

$$y_3 = 140 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 59.70$$

$$\therefore \text{Centroid} = (0, 59.70)$$

(iii)



Solⁿ:

$$A_1 = 12 \times 188 = 2256 \text{ mm}^2$$

$$A_2 = 12 \times 150 = 1800 \text{ mm}^2$$

$$x_1 = 6, x_2 = 75$$

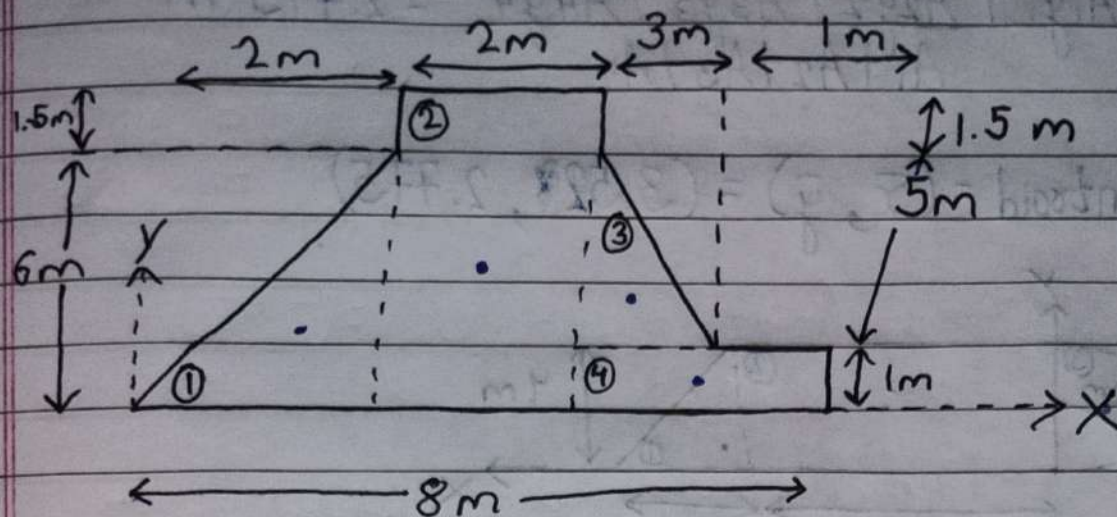
$$y_1 = 94, y_2 = 194$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = 36.62$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = 138.37$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (36.62, 138.37).$$

(iv)

Solⁿ:

$$A_1 = \frac{1}{2} \times 2 \times 6 = 6 \text{ m}^2$$

$$A_2 = 2 \times 7.5 = 15 \text{ m}^2$$

$$A_3 = \frac{1}{2} \times 3 \times 5 = 7.5 \text{ m}^2$$

$$A_4 = 1 \times 4 = 4 \text{ m}^2$$

$$\cancel{y_1} = 2 \text{ m}, \cancel{x_1} = 1.33 \text{ m}$$

$$x_2 = 3 \text{ m}, y_2 = 3.75 \text{ m}$$

$$x_3 = 5.5 \text{ m}, y_3 = 2.66 \text{ m}$$

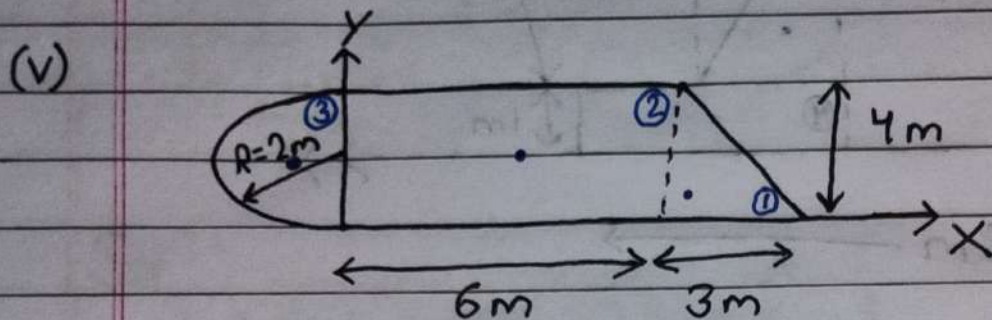
$$x_4 = 6 \text{ m}, y_4 = 0.5 \text{ m}$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4}{A_1 + A_2 + A_3 + A_4} = \frac{6 \times 1.33 + 15 \times 3 + 7.5 \times 5.5 + 4 \times 6}{6 + 15 + 7.5 + 4}$$

$$= 3.522 \text{ m}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3 + A_4 y_4}{A_1 + A_2 + A_3 + A_4} = 2.775 \text{ m}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (3.521, 2.775)$$



Solⁿ:

$$A_1 = \frac{1}{2} \times 3 \times 4 = 6 \text{ m}^2$$

$$A_2 = 6 \times 4 = 24 \text{ m}^2$$

$$A_3 = \frac{\pi R^2}{2} = \frac{\pi \times 2^2}{2} = 2\pi = 6.283 \text{ m}^2$$

$$x_1 = 6 + \frac{1}{3} \times 3 = 7 \text{ m}, \quad y_1 = \frac{4}{3} = 1.33 \text{ m}$$

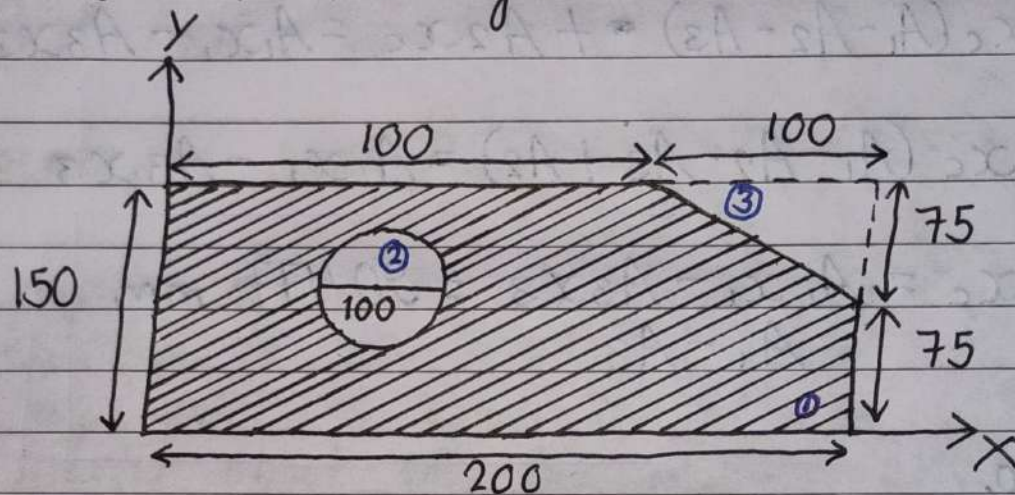
$$x_2 = 3 \text{ m}, \quad y_2 = 2 \text{ m}$$

$$x_3 = -\frac{4R}{3\pi} = -0.848 \text{ m}, \quad y_3 = 2 \text{ m}$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3}{A_1 + A_2 + A_3} = 2.995 \text{ m}, \quad \bar{y} = 1.889 \text{ m}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (2.995, 1.889)$$

Imp 9. Determine the coordinates x_c & y_c of the centre of a 100 mm diameter circular hole cut in a thin plate, so that this point will be the centroid of the remaining shaded area as shown in the figure. (All dimensions are in mm)



Solⁿ:

$$A_1 = 200 \times 150 = 30000 \text{ mm}^2$$

$$A_2 = \pi r^2 = \pi \times \left(\frac{100}{2}\right)^2 = 7853.98 \text{ mm}^2$$

$$A_3 = \frac{1}{2} \times 75 \times 100 = 3750 \text{ mm}^2$$

$$x_1 = 100 \text{ mm}, y_1 = 75 \text{ mm}$$

$$x_2 = x_c, y_2 = y_c$$

$$x_3 = \frac{2}{3} \times 100 + 100 = 166.66 \text{ mm}$$

$$y_3 = \frac{2}{3} \times 75 + 75 = 125 \text{ mm}$$

$$\bar{x} = \frac{A_1 x_1 - A_2 x_2 - A_3 x_3}{A_1 - A_2 - A_3}$$

$$\Rightarrow x_c (A_1 - A_2 - A_3) = A_1 x_1 - A_2 x_c - A_3 x_3$$

$$\Rightarrow x_c (A_1 - A_2 - A_3) + A_2 x_c = A_1 x_1 - A_3 x_3$$

$$\Rightarrow x_c (A_1 - A_2 - A_3 + A_2) = A_1 x_1 - A_3 x_3$$

$$\Rightarrow x_c = \frac{A_1 x_1 - A_3 x_3}{A_1 - A_3} = 90.477 \text{ mm}$$

Also,

$$y_c = \frac{A_1 y_1 - A_3 y_3}{A_1 - A_3} = 67.857 \text{ mm}$$

Hence, the centroid of the shaded area is,

$$\text{Centroid} = (\bar{x}, \bar{y}) = (x_c, y_c) = (90.477, 67.857).$$

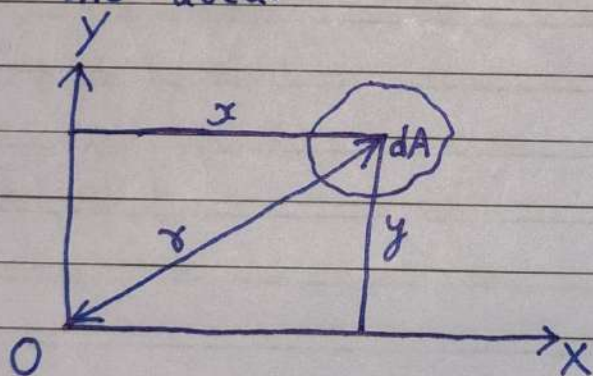
Theorems of Moments of Inertia

There are two theorems of Moment of Inertia:

1. Perpendicular Axis Theorem
2. Parallel Axis Theorem

Perpendicular Axis Theorem

The moment of inertia of an area about an axis perpendicular to its plane (polar moment of inertia) at any point is equal to the sum of moments of inertia about any two mutually perpendicular axes through the same point and lying in the plane of the area.



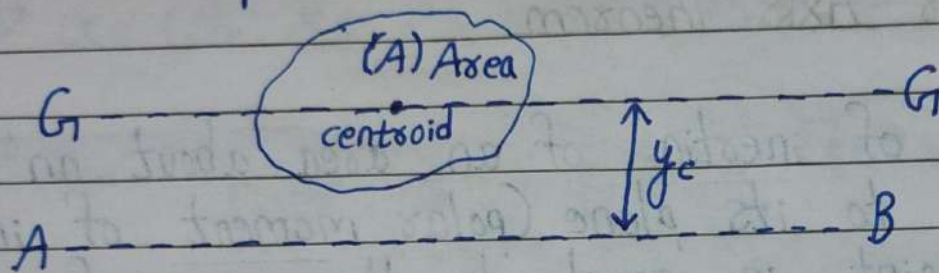
$$I_{zz} = I_{xx} + I_{yy}$$

$$[I = Ax^2]$$

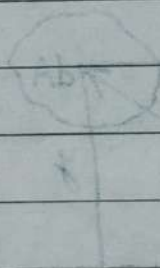
$$\begin{aligned} I_{zz} &= r^2 \cdot dA = (x^2 + y^2) dA \\ \Rightarrow I_{zz} &= (x^2 + y^2) A \\ \Rightarrow I_{zz} &= x^2 \cdot A + y^2 \cdot A \\ \Rightarrow I_{zz} &= I_{yy} + I_{xx} \end{aligned}$$

Parallel Axis Theorem

Moment of inertia about any axis in the plane of an area is equal to the sum of moment of inertia about a parallel centroidal axis and the product of area and square of the distance between the two parallel axis.



$$I_{AB} = I_{GG} + A \cdot y_c^2$$



$$I_x A = T$$

$$I_y T + x x I = r r I$$

$$A b (r^2 y + x^2 x) = A b r^2 = r r T$$

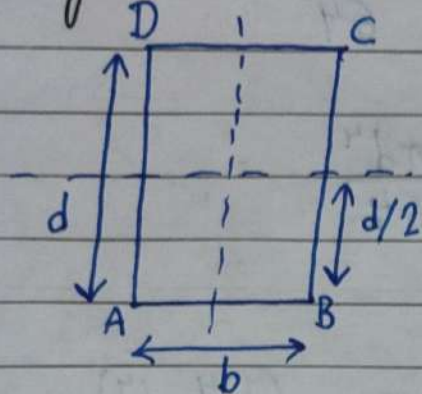
$$A (r^2 y + x^2 x) = r r T \Leftarrow$$

$$A r^2 y + A x^2 x = r r T \Leftarrow$$

$$x x T + y y T = r r T \Leftarrow$$

Moment of Inertia of Standard Sections

1. Rectangle

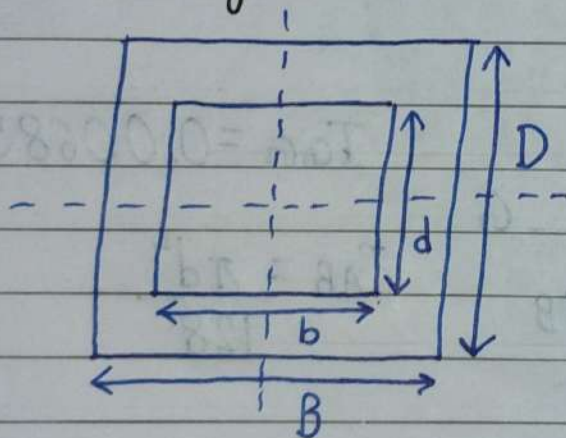


$$I_{xx} = \frac{bd^3}{12}$$

$$I_{yy} = \frac{db^3}{12}$$

$$I_{AB} = I_{GG} + A \cdot y_c^2 = \frac{bd^3}{12} + b \cdot d \cdot \left(\frac{d}{2}\right)^2 = \frac{bd^3}{3}$$

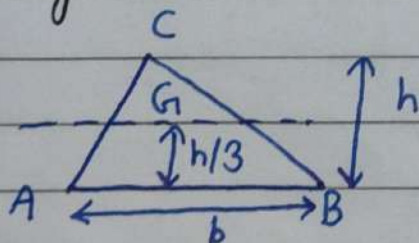
2. Hollow Rectangle



$$I_{xx} = \frac{BD^3 - bd^3}{12}$$

$$I_{yy} = \frac{DB^3 - db^3}{12}$$

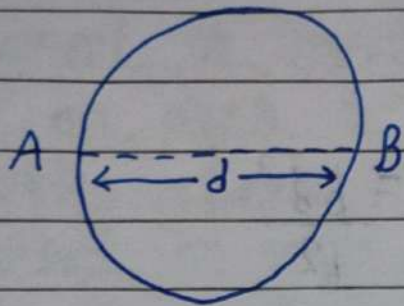
3. Triangle



$$I_{xx} = \frac{bh^3}{36}$$

$$I_{AB} = I_{GG} + A \cdot y_c^2 = \frac{bh^3}{36} + \frac{1}{2} \times b \times h \times \left(\frac{h}{3}\right)^2 = \frac{bh^3}{12}$$

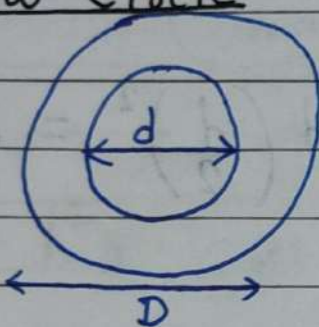
4. Circle



$$I_{xx} = \frac{\pi d^4}{64}$$

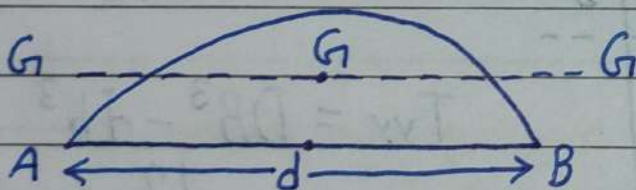
$$I_{yy} = \frac{\pi d^4}{64}$$

5. Hollow Circle



$$I = \frac{\pi}{64} (D^4 - d^4)$$

Semi-circle

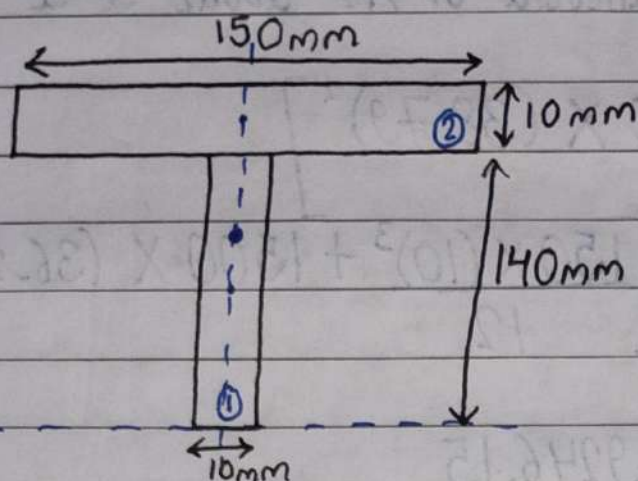


$$I_{GG} = 0.00685 d^4$$

$$I_{AB} = \frac{\pi d^4}{128}$$

Numericals

Q.



Determine the moment of inertia of the section shown in figure about an axis passing through the centroid and parallel to the top-most fibre of the section? Also determine moment of inertia about the axis of symmetry. Hence find radii of ~~direction~~ gyration.

Solⁿ:

$$A_1 = 10 \times 140 = 1400 \text{ mm}^2$$

$$A_2 = 10 \times 150 = 1500 \text{ mm}^2$$

$\bar{x}_c = 0$, since the given section is symmetric to Y-axis.

$$y_1 = 70 \text{ mm}, y_2 = 140 + 5 = 145 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = 108.793 \text{ mm}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (0, 108.793) \text{ [From Bottom]} \\ = (0, 41.207) \text{ [From Top]}$$

I_{xx} = Moment of Inertia of A_1 about $x-x$ axis
+ Moment of Inertia of A_2 about $x-x$ axis

$$= \left[\frac{10 \times (140)^3}{12} + 1400 \times (38.79)^2 \right] + \left[\frac{150 \times (10)^3}{12} + 1500 \times (36.21)^2 \right]$$

$$= 4393196.41 + 1979246.15$$

$$\Rightarrow I_{xx} = 6,372,442.56 \text{ mm}^4$$

Now,

$$I_{yy} = \frac{140 \times (10)^3}{12} + \frac{10 \times (150)^3}{12} = 2824166.666 \text{ mm}^4$$

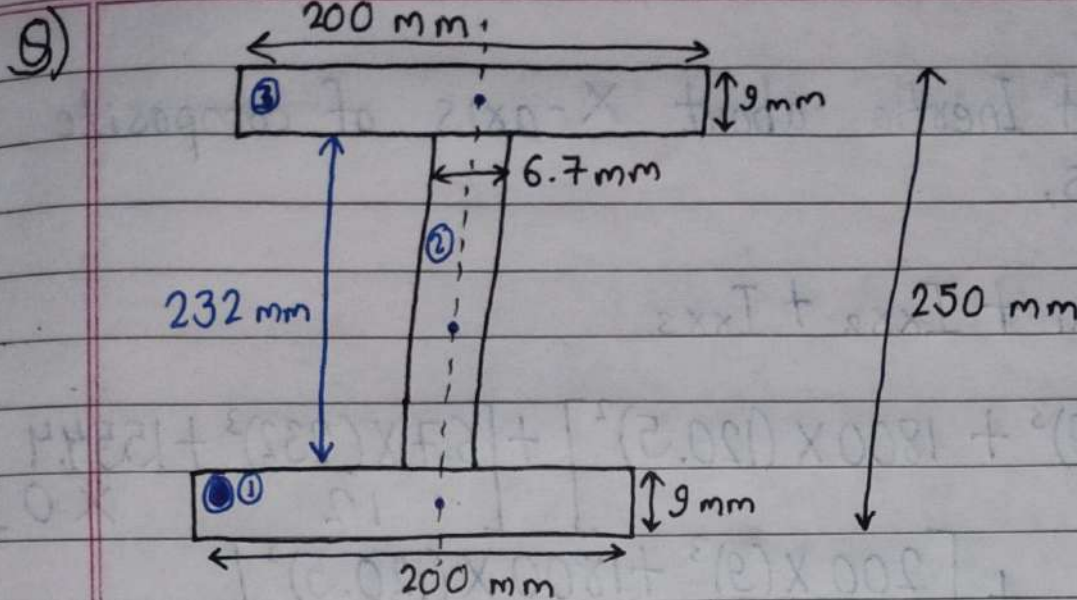
We have,

$$I = AK^2$$
$$\Rightarrow K = \sqrt{\frac{I}{A}}$$

So,

$$K_{xx} = \sqrt{\frac{I_{xx}}{A}} = 46.876 \text{ mm}$$

$$\& K_{yy} = \sqrt{\frac{I_{yy}}{A}} = 31.206 \text{ mm}$$



Determine the moment of inertia of symmetric I-section shown in figure about its centroidal axis XX & YY?

Solⁿ:

$$A_1 = 200 \times 9 = 1800 \text{ mm}^2$$

$$A_2 = 232 \times 6.7 = 1554.4 \text{ mm}^2$$

$$A_3 = A_1 = 1800 \text{ mm}^2$$

$$\bar{x} = 0$$

$$y_1 = 4.5 \text{ mm}, y_2 = 116 + 9 = 125 \text{ mm}, y_3 = 4.5 + 241 = 245.5 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 125 \text{ mm}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (0, 125)$$

Now,
Moment of Inertia about X-axis of composite section is,

$$I_{xx} = I_{xx1} + I_{xx2} + I_{xx3}$$

$$= \left[\frac{200 \times (9)^3}{12} + 1800 \times (120.5)^2 \right] + \left[\frac{6.7 \times (232)^3}{12} + 1554.4 \times 0 \right] + \left[\frac{200 \times (9)^3}{12} + 1800 \times (120.5)^2 \right]$$

$$= 26148600 + 6972002.1333 + 26148600$$

$$\Rightarrow I_{xx} = 59,269,202.1333 \text{ mm}^4.$$

Similarly,

$$I_{yy} = I_{yy1} + I_{yy2} + I_{yy3}$$

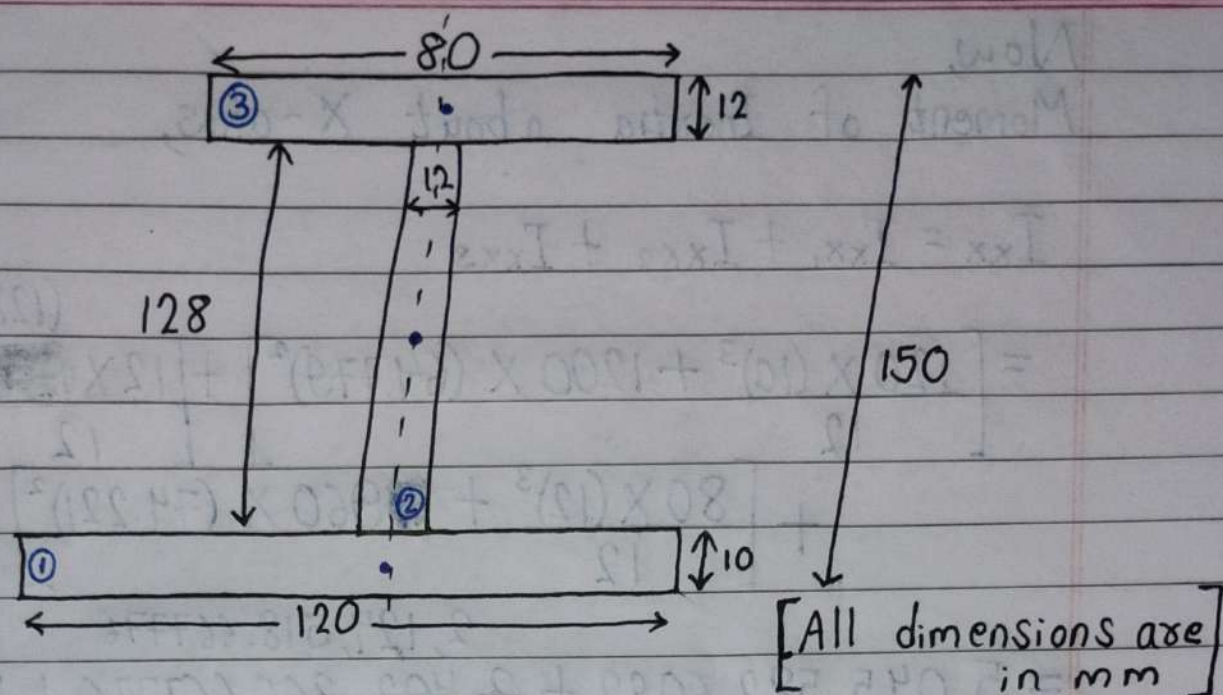
$$= \frac{9 \times (200)^3}{12} + 232 \times (6.7)^3 + \frac{9 \times (200)^3}{12}$$

$$= 12,005,814.75133 \text{ mm}^4.$$

Also, the polar moment of inertia is,

$$I_{zz} = I_{xx} + I_{yy} = 71,275,016.884 \text{ mm}^4.$$

9)



Determine the moment of inertia of I-section shown in figure above - and also polar moment of inertia?

Solⁿ:

$$A_1 = 120 \times 10 = 1200 \text{ mm}^2$$

$$A_2 = 128 \times 12 = 1536 \text{ mm}^2$$

$$A_3 = 80 \times 12 = 960 \text{ mm}^2$$

$$\bar{x} = 0$$

$$y_1 = 5 \text{ mm}, y_2 = 10 + 64 = 74 \text{ mm} \text{ \& } y_3 = 144 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 69.779 \text{ mm}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (0, 69.779)$$

Now,

Moment of Inertia about X-axis,

$$I_{xx} = I_{xx1} + I_{xx2} + I_{xx3}$$

$$= \left[\frac{120 \times (10)^3}{12} + 1200 \times (64.779)^2 \right] + \left[\frac{12 \times (128)^3}{12} + 1536 \times (4.221)^2 \right] + \left[\frac{80 \times (12)^3}{12} + 960 \times (74.221)^2 \right]$$

$$= 5,045,582.6092 + 2,124,518.667776 + 5,299,926.56736$$

$$\Rightarrow I_{xx} = 13,747,875.844336 \text{ mm}^4$$

$$\Rightarrow I_{xx} = 12,470,027.844 \text{ mm}^4$$

Also,

$$I_{yy} = I_{yy1} + I_{yy2} + I_{yy3}$$

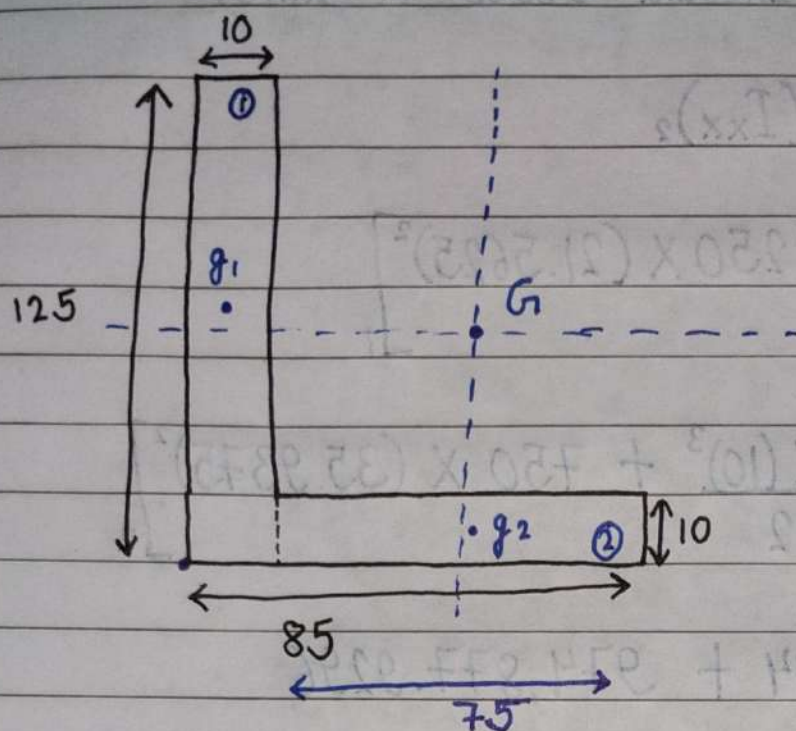
$$= \left[\frac{10 \times (120)^3}{12} \right] + \left[\frac{1280 \times (12)^3}{12} \right] + \left[\frac{12 \times (80)^3}{12} \right]$$

$$= 1,440,000 + 18,432 + 512,000$$

$$\Rightarrow I_{yy} = 1,970,432 \text{ mm}^4$$

$$\therefore I_{zz} = I_{xx} + I_{yy} = 14,440,459.844 \text{ mm}^4$$

#



Determine the moment of inertia of given L-section?

Solⁿ:

$$A_1 = 125 \times 10 = 1250 \text{ mm}^2$$

$$A_2 = 75 \times 10 = 750 \text{ mm}^2$$

$$x_1 = 5 \text{ mm}, y_1 = 62.5 \text{ mm}$$

$$x_2 = 7.5 \text{ mm}, y_2 = 5 \text{ mm}$$

$$\bar{x} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = 20.9375 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = 40.9375 \text{ mm}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (20.9375, 40.9375)$$

Now, Moment of Inertia about X-axis,

$$I_{xx} = (I_{xx})_1 + (I_{xx})_2$$

$$= \left[\frac{10 \times (125)^3}{12} + 1250 \times (21.5625)^2 \right]_1$$

$$+ \left[\frac{75 \times (10)^3}{12} + 750 \times (35.9375)^2 \right]_2$$

$$= 2,208,780.9244 + 974,877.9296$$

$$\Rightarrow I_{xx} = 3,183,658.854 \text{ mm}^4$$

Also, moment of inertia about Y-axis,

$$I_{yy} = (I_{yy})_1 + (I_{yy})_2$$

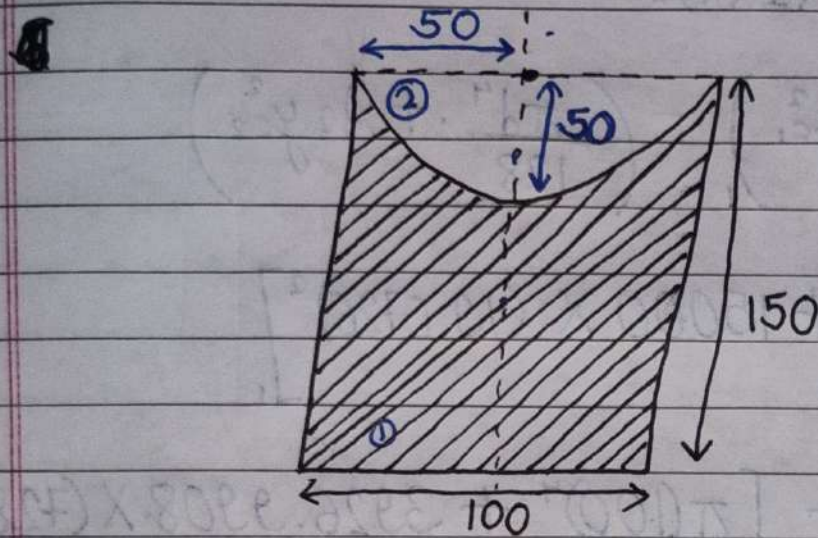
$$= \left[\frac{125 \times (10)^3}{12} + 1250 \times (15.9375)^2 \right]_1$$

$$+ \left[\frac{10 \times (75)^3}{12} + 750 \times (26.5625)^2 \right]_2$$

$$I_{yy} = 1,208,658.854 \text{ mm}^4$$

$$\text{Polar moment of inertia, } I_{zz} = I_{xx} + I_{yy} = 4,392,317.708 \text{ mm}^4$$

Q) Determine the moment of inertia of given figure and also the polar moment of inertia?



Solⁿ:

$$A_1 = 150 \times 100 = 15000 \text{ mm}^2$$

$$A_2 = \frac{\pi r^2}{2} = \frac{\pi \times (50)^2}{2} = 3926.9908 \text{ mm}^2$$

$$\bar{x}_2 = 50 \text{ mm}$$

$$y_1 = 75 \text{ mm} \quad \& \quad y_2 = 150 - \frac{4r}{3\pi} = 128.7793 \text{ mm}$$

$$\bar{y} = \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2} = 55.9274 \text{ mm}$$

$$\therefore \text{Centroid} = (\bar{x}, \bar{y}) = (50, 55.9274)$$

Now, the moment of inertia about X-axis is,

$$I_{xx} = (I_{xx})_1 - (I_{xx})_2$$

$$= \left(\frac{bd^3}{12} + A_1 y_{c1}^2 \right)_1 - \left(\frac{\pi d^4}{128} + A_2 y_{c2}^2 \right)_2$$

$$= \left[\frac{100 \times (150)^3}{12} + 15000 \times (19.0726)^2 \right]_1$$

$$- \left[\frac{0.00685 \times (100)^4}{128} + 3926.9908 \times (72.8519)^2 \right]_2$$

$$= 33,581,461.0614 - 21,527,108.3550$$

$$\cancel{I_{xx} = 10,284,983.4458 \text{ mm}^4}$$

$$\Rightarrow I_{xx} = 12,054,352.7064 \text{ mm}^4$$

The moment of inertia about Y-axis is,

$$I_{yy} = (I_{yy})_1 - (I_{yy})_2$$

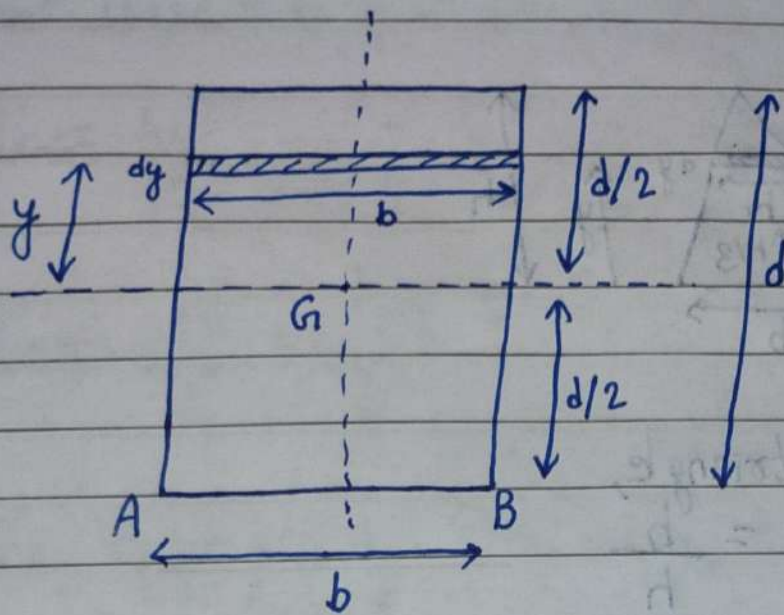
$$= \frac{150 \times (100)^3}{12} - \frac{\pi \times (100)^4}{128}$$

$$= 12,500,000 - 2,454,369.2606$$

$$\Rightarrow I_{yy} = 10,045,630.7394 \text{ mm}^4$$

$$\therefore I_{zz} = I_{xx} + I_{yy} = 22,099,983.4458 \text{ mm}^4$$

Moment of Inertia of a Rectangle about its Centroidal Axis



$$I_{\text{strip}} = b \cdot dy \cdot y^2$$

$$I_{\text{Rectangle}} = \int_{-d/2}^{+d/2} I_{\text{strip}} = \int_{-d/2}^{+d/2} b \cdot dy \cdot y^2 = b \cdot \left[\frac{y^3}{3} \right]_{-d/2}^{+d/2} = \frac{bd^3}{12}$$

$$\Rightarrow I_{xx} = \frac{bd^3}{12} = I_{GG}$$

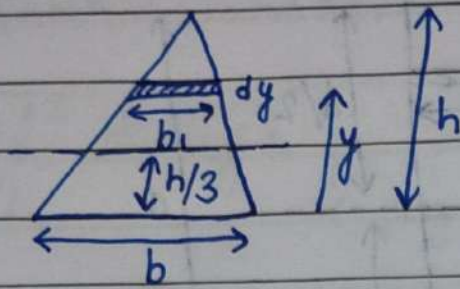
Similarly,

$$I_{yy} = \frac{db^3}{12}$$

$$I_{AB} = I_{GG} + A \cdot y_c^2 = \frac{bd^3}{12} + b \cdot d \left(\frac{d}{2} \right)^2$$

$$\Rightarrow I_{AB} = \frac{bd^3}{3}$$

Moment of Inertia of a ~~triangle~~^{Tri-}angle About its Centroidal Axis



We know for triangle,
$$\frac{b_1}{(h-y)} = \frac{b}{h}$$

$$\Rightarrow b_1 = \frac{b}{h} \cdot (h-y)$$

Now,

$$I_{\text{strip}} = b_1 \cdot dy \cdot y^2$$

$$I_b = \int_0^h I_{\text{strip}} = \int_0^h \frac{b}{h} (h-y) y^2 dy$$

$$\Rightarrow I_b = \frac{b}{h} \int_0^h (hy^2 - y^3) dy$$

$$\Rightarrow I_b = \frac{b}{h} \left[\frac{hy^3}{3} - \frac{y^4}{4} \right]_0^h$$

$$\Rightarrow I_b = \frac{b}{h} \left[+\frac{h^4}{3} - \frac{h^4}{4} \right]$$

$$\Rightarrow I_b = \frac{bh^3}{12} \quad [\text{About Base}]$$

Then,

$$I_{GG} = I_{base} - Ay^2$$

$$\Rightarrow I_{GG} = \frac{bh^3}{12} - \left(\frac{1}{2}bh\right) \cdot \left(\frac{h}{3}\right)^2$$

$$\Rightarrow I_{GG} = \frac{bh^3}{12} - \frac{bh^3}{18}$$

$$\Rightarrow I_{GG} = \frac{3bh^3 - 2bh^3}{36}$$

$$\Rightarrow \boxed{I_{GG} = \frac{bh^3}{36}}$$

[About Centroidal Axis]